

# TCPDUMP & LIBPCAP

A quick-reference guide for tcpdump packet capture: command-line options, BPF filter syntax, and TCP/IP header layout for traffic analysis.

```
tcpdump [-aAenStvxX] [-F filterfile] [-i interface] [-c count] [-r pcapfile] [-s snaplen] [-w pcapfile] ['bpf filter']
```

## tcpdump Usage

1

| OPTION          | MEANING                                      |
|-----------------|----------------------------------------------|
| -A              | Display each packet's payload in ASCII.      |
| -c n            | Exit after capturing n packets.              |
| -D              | List available network interfaces.           |
| -e              | Print the data link layer (Ethernet) header. |
| -F file         | Read the BPF filter expression from a file.  |
| -i iface        | Listen on the specified interface.           |
| -n              | Do not resolve IP addresses or port names.   |
| -r file         | Read packets from a capture file.            |
| -s snaplen      | Set the snapshot length in bytes.            |
| -S              | Print absolute TCP sequence numbers.         |
| -t              | Do not print a timestamp on each line.       |
| -tttt           | Print a date-and-time timestamp.             |
| -v / -vv / -vvv | Increase verbosity (more v = more detail).   |
| -w file         | Write raw packets to a pcap file.            |
| -X              | Print each packet in hex.                    |
| -xx             | Print the link layer header in hex too.      |
| -X              | Print payload in hex <i>and</i> ASCII.       |

## Useful One-Liners

2

| TASK                   | COMMAND                                                  |
|------------------------|----------------------------------------------------------|
| Capture an interface   | tcpdump -i eth0                                          |
| Stop name resolution   | tcpdump -i eth0 -n                                       |
| Capture 100 packets    | tcpdump -i eth0 -c 100                                   |
| Save to a pcap file    | tcpdump -i eth0 -w capture.pcap                          |
| Read a pcap file       | tcpdump -r capture.pcap                                  |
| Traffic to/from a host | tcpdump -i eth0 host 10.0.0.1                            |
| HTTP (port 80)         | tcpdump -i eth0 tcp port 80                              |
| HTTPS / TLS (443)      | tcpdump -i eth0 tcp port 443                             |
| DNS (UDP 53)           | tcpdump -i eth0 udp port 53                              |
| SYN packets            | tcpdump -i eth0 'tcp[13] & 2 != 0'                       |
| SYN-ACK packets        | tcpdump -i eth0 'tcp[13] & 2 != 0 and tcp[13] & 16 != 0' |
| RST packets            | tcpdump -i eth0 'tcp[13] & 4 != 0'                       |
| ICMP (ping)            | tcpdump -i eth0 icmp                                     |
| ARP traffic            | tcpdump -i eth0 arp                                      |
| Hex + ASCII view       | tcpdump -i eth0 -X -n                                    |
| Exclude a host         | tcpdump -i eth0 not host 10.0.0.1                        |
| Verbose capture        | tcpdump -i eth0 -vvv -n                                  |

## ICMP Types

3

| TYPE / CODE | NAME                               |
|-------------|------------------------------------|
| 0 / 0       | Echo Reply                         |
| 3 / 0       | Destination Network Unreachable    |
| 3 / 1       | Destination Host Unreachable       |
| 3 / 2       | Protocol Unreachable               |
| 3 / 3       | Port Unreachable                   |
| 3 / 4       | Fragmentation Required             |
| 4 / 0       | Source Quench                      |
| 5 / 0       | Network Redirect                   |
| 5 / 1       | Host Redirect                      |
| 8 / 0       | Echo Request (ping)                |
| 9 / 0       | Router Advertisement               |
| 11 / 0      | Time to Live Exceeded (in transit) |
| 11 / 1      | Fragment Reassembly Time Exceeded  |
| 12 / 0      | Parameter Problem                  |
| 13 / 0      | Timestamp Request                  |
| 17 / 0      | Address Mask Request               |

## BPF Filter Syntax

| ELEMENT             | MEANING                    | EXAMPLE                              |
|---------------------|----------------------------|--------------------------------------|
| host                | Match a host               | host 10.0.0.1                        |
| src / dst           | Direction qualifiers       | src host 10.0.0.1, dst host 10.0.0.2 |
| net                 | Match a network            | net 192.168.1.0/24                   |
| port                | Match a port               | port 80                              |
| src port / dst port | Match source/dest port     | src port 443                         |
| portrange           | Match a port range         | portrange 6000-6010                  |
| proto               | Match a protocol           | tcp, udp, icmp, ip, arp              |
| and / &&            | Both conditions            | host 10.0.0.1 and tcp port 80        |
| or /                | Either condition           | tcp port 80 or udp port 53           |
| not / !             | Negation                   | tcp and not port 22                  |
| greater / less      | Match packet length        | less 300                             |
| ip[0], ip[1], ...   | Byte-level tests           | tcp[13] & 2 != 0                     |
| src net / dst net   | Source/destination network | src net                              |
| ( )                 | Grouping                   | (tcp or udp) and not port 53         |

Byte-level tests use offsets: ip[9] (protocol), tcp[13] (flags), tcp[13]&2 != 0 (SYN set).

4

## DNS Header & Flags

5

| FIELD            | NOTES                                         |
|------------------|-----------------------------------------------|
| Query ID         | 2 bytes — matches a query to its response.    |
| Flags            | QR, Opcode, AA, TC, RD, RA, Z, AD, CD, RCODE. |
| Query Count      | Number of questions.                          |
| Answer Count     | Number of answers.                            |
| Authority Count  | Number of authority records.                  |
| Additional Count | Number of additional records.                 |

### UDP Header

| FIELD            | BYTES | NOTES                                          |
|------------------|-------|------------------------------------------------|
| Source Port      | 2     | Source service port                            |
| Destination Port | 2     | Destination service port                       |
| Length           | 2     | Includes the 8-byte header; minimum value is 8 |
| Checksum         | 2     | Covers pseudo-header, UDP header and payload   |

6

## IPv4 Header

| FIELD               | OFFSET         | NOTES                                          |
|---------------------|----------------|------------------------------------------------|
| Version             | ip[0]&0xf0     | 4 = IPv4                                       |
| Header Length (IHL) | ip[0]&0xf      | In 4-byte words; minimum 5 (20 bytes)          |
| TOS / DSCP          | ip[1]          | Differentiated Services                        |
| Total Length        | ip[2:2]        | Includes header                                |
| Identification      | ip[4:2]        | Unique ID for fragmented packets               |
| Flags (DF / MF)     | ip[6]          | X reserved, D don't fragment, M more fragments |
| Fragment Offset     | ip[6:2]&0x1fff | Multiply by 8                                  |
| TTL                 | ip[8]          | Number of hops before packet is discarded      |
| Protocol            | ip[9]          | 1 ICMP, 6 TCP, 17 UDP, 47 GRE, 50 ESP, 51 AH   |
| Header Checksum     | ip[10:2]       | IP header only                                 |
| Source Address      | ip[12:4]       | Initial sender's IP address                    |
| Destination Address | ip[16:4]       | Receiving host's IP address                    |
| Options             | ip[20..]       | Up to 40 bytes                                 |

7

## TCP Header

8

| FIELD          | OFFSET      | NOTES                                                |
|----------------|-------------|------------------------------------------------------|
| Source Port    | tcp[0:2]    |                                                      |
| Dest Port      | tcp[2:2]    |                                                      |
| Seq Number     | tcp[4:4]    | Increments each byte sent                            |
| Ack Number     | tcp[8:4]    | Next expected sequence #                             |
| Header Len     | tcp[12]>>4  | In 4-byte words; min 5                               |
| Reserved       | tcp[12]&0xf | Set to 0                                             |
| Flags          | tcp[13]     | CWR ECE URG ACK PSH RST SYN FIN                      |
| Window Size    | tcp[14:2]   | Receive window size                                  |
| Checksum       | tcp[16:2]   | Covers pseudo-header + TCP header + payload          |
| Urgent Pointer | tcp[18:2]   |                                                      |
| Options        | tcp[20..]   | Up to 40 bytes (MSS, window scale, SACK, timestamps) |

## TCP Flags

9

| SINGLE FLAGS |      |                    | FLAG COMBINATIONS |      |                  | SCAN PATTERNS |      |                 |
|--------------|------|--------------------|-------------------|------|------------------|---------------|------|-----------------|
| FLAG         | HEX  | DESC               | FLAGS             | HEX  | DESC             | PATTERN       | HEX  | DESC            |
| FIN          | 0x01 | Finish             | SYN-ACK           | 0x12 | Handshake reply  | NULL          | 0x00 | No flags        |
| SYN          | 0x02 | Synchronize        | FIN-ACK           | 0x11 | Close + ack      | SYN           | 0x02 | Half-open scan  |
| RST          | 0x04 | Reset              | RST-ACK           | 0x14 | Abort + ack      | FIN           | 0x01 | FIN scan        |
| PSH          | 0x08 | Push               | PSH-ACK           | 0x18 | Push + ack       | XMAS          | 0x29 | FIN+PSH+URG     |
| ACK          | 0x10 | Acknowledgment     | SYN-ACK-PSH       | 0x1A | Handshake + push | ACK           | 0x10 | ACK probe       |
| URG          | 0x20 | Urgent             | URG-ACK           | 0x3A | Urgent + ack     | ALL           | 0xFF | All flags set   |
| ECE          | 0x40 | ECN-Echo           | ECE-CWR           | 0xC0 | ECN congestion   | MAIMON        | 0x11 | FIN+ACK scan    |
| CWR          | 0x80 | Congestion Reduced | ACK               | 0x10 | Ack only         | WINDOW        | 0x10 | ACK window scan |
|              |      |                    | SYN-FIN           | 0x03 | Inconsistent     | SUPER-XMAS    | 0x38 | XMAS + SYN+ACK  |
|              |      |                    | SYN-RST           | 0x06 | Inconsistent     |               |      |                 |
|              |      |                    | FIN-PSH           | 0x09 | Close + push     |               |      |                 |
|              |      |                    | FIN-URG           | 0x21 | Close + urgent   |               |      |                 |

## Common Ports & Acronyms

10

| TCP  |          | UDP       |                        | ACRONYMS |                                                         |
|------|----------|-----------|------------------------|----------|---------------------------------------------------------|
| PORT | SERVICE  | PORT      | SERVICE                | ACRONYM  | MEANING                                                 |
| 20   | ftp-data | 7         | echo                   | AH       | Authentication Header (RFC 2402)                        |
| 21   | FTP      | 19        | chargen                | ARP      | Address Resolution Protocol (RFC 826)                   |
| 22   | SSH      | 53        | domain (DNS)           | BGP      | Border Gateway Protocol (RFC 1771)                      |
| 23   | Telnet   | 67 / 68   | DHCP server / client   | CWR      | Congestion Window Reduced (RFC 2481)                    |
| 25   | SMTP     | 69        | TFTP                   | DF       | Do Not Fragment flag (RFC 791)                          |
| 43   | WHOIS    | 123       | NTP                    | DHCP     | Dynamic Host Configuration Protocol (RFC 2131)          |
| 53   | DNS      | 137       | netbios-ns             | DNS      | Domain Name System (RFC 1035)                           |
| 80   | HTTP     | 138       | netbios-dgm            | ECN      | Explicit Congestion Notification (RFC 3168)             |
| 88   | Kerberos | 161       | SNMP                   | ESP      | Encapsulating Security Payload (RFC 2406)               |
| 110  | POP3     | 162       | SNMP-trap              | FTP      | File Transfer Protocol (RFC 959)                        |
| 113  | authd    | 500       | ISAKMP                 | GRE      | Generic Route Encapsulation (RFC 2784)                  |
| 119  | NNTP     | 514       | syslog                 | HTTP     | Hypertext Transfer Protocol (RFC 1945)                  |
| 143  | IMAP     | 520       | RIP                    | ICMP     | Internet Control Message Protocol (RFC 792)             |
| 179  | BGP      | 546 / 547 | DHCPv6 client / server | IGMP     | Internet Group Management Protocol (RFC 2236)           |
| 443  | HTTPS    | 1900      | SSDP                   | IMAP     | Internet Message Access Protocol (RFC 2060)             |
| 445  | MS-SMB   | 5353      | mDNS                   | IP       | Internet Protocol (RFC 791)                             |
| 465  | SMTPS    |           |                        | ISAKMP   | Internet Security Assoc. & Key Mgmt Protocol (RFC 7296) |
| 1433 | MS-SQL   |           |                        | L2TP     | Layer 2 Tunneling Protocol (RFC 2661)                   |
| 3128 | Squid    |           |                        | OSPF     | Open Shortest Path First (RFC 1583)                     |
| 3306 | MySQL    |           |                        | POP3     | Post Office Protocol v3 (RFC 1460)                      |
| 3389 | RDP      |           |                        | RFC      | Request for Comments                                    |
|      |          |           |                        | SMTP     | Simple Mail Transfer Protocol (RFC 821)                 |
|      |          |           |                        | SSH      | Secure Shell (RFC 4253)                                 |
|      |          |           |                        | SSL      | Secure Sockets Layer (RFC 6101)                         |
|      |          |           |                        | TCP      | Transmission Control Protocol (RFC 793)                 |
|      |          |           |                        | TLS      | Transport Layer Security (RFC 5246)                     |
|      |          |           |                        | TFTP     | Trivial File Transfer Protocol (RFC 1350)               |
|      |          |           |                        | TOS      | Type of Service (RFC 2474)                              |
|      |          |           |                        | UDP      | User Datagram Protocol (RFC 768)                        |